

73-103 · CMU-Q

Principles of Macroeconomics

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Andrés Castaño Zuluaga
Carnegie Mellon University Qatar
Business Administration

Labor Markets

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Course roadmap

	Topic	Description	Toolkit	In the world
Measurement	GDP	What the economy produced	National accounts; $Y = C + I + G + NX$	Read a GDP release; split oil vs non-oil
	Prices	Persistent rise in the price level	CPI and GDP deflator; real vs nominal	Index a wage; judge a currency peg
	Labor	Employment, unemployment, participation	Labor-force identities; u and LFPR	Read a labor survey; GCC hiring rules
Short-run demand	Goods market	What determines demand? Equilibrium and the multiplier.	Goods-market equilibrium; multiplier	Size a fiscal package after an oil shock
	Financial markets	How i is set	Money market; the policy-rate rule	Read a QCB or Fed rate decision
	IS-LM	Goods + financial together	IS and LM together; the policy mix	Map a shock onto Y and the interest rate
Medium run	Phillips	Inflation and unemployment together	Short-run Phillips curve	Trade off inflation vs jobs in a boom
	Growth	Why levels differ	Production function; growth accounting	Explain why living standards differ
Open	Open economy	Trade, rates, and the peg	Net exports; peg vs float	Trace a shock through NX and the riyal peg

Midterm · Debate

Class roadmap

1 Discussion

Article review and class discussion

2 Measuring Unemployment

Key labor market indicators

3 Types of Unemployment

Frictional, structural, cyclical

4 Active Learning 1

Labor force calculations

5 Why Unemployment Exists

Labor supply and demand

6 Okun's Law & Phillips Curve

Output-unemployment-inflation links

7 Active Learning 2

Unemployment concepts

8 AI & Labor Markets

Technology and the future of work

9 Key Takeaways

Summary and conclusions

SECTION 01

Discussion

Labor Market Indicators Point in All Directions

Peterson Institute for International Economics (PIIE) | January 16, 2026

The U.S. labor market currently presents a puzzle: some indicators are at their strongest levels in 25 years, while others suggest weakness rarely seen outside recessions.

Indicator	Current Status	What It Reveals
Prime-Age Emp-Pop Ratio	Historically strong	More working-age adults have jobs than in decades
Real Wages	Near all-time highs	Workers' purchasing power is increasing
Hiring Rate	Lowest since 2012	Employers are reluctant to add new workers
Monthly Payroll Growth	Weak	Job creation has slowed significantly

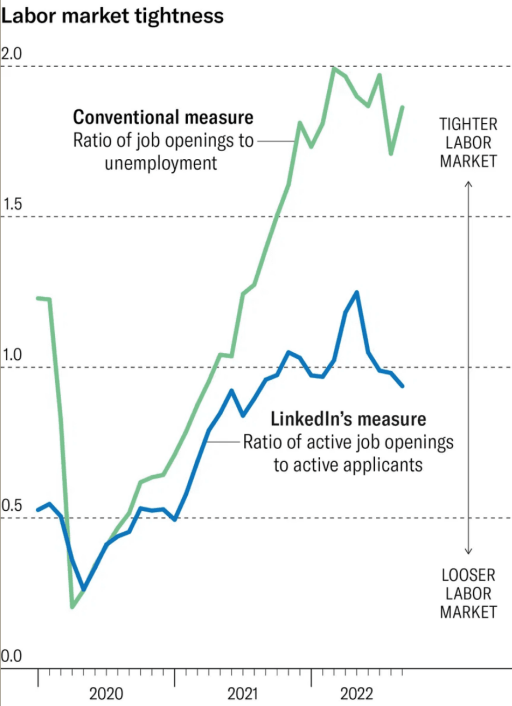
Key Insight

No single indicator tells the complete story. The unemployment rate alone can mask important dynamics in the labor market.

[Read the article here](#)

How Tight Is the U.S. Labor Market?

LinkedIn's new measure of labor market slack suggests the labor market is not as tight as traditional measures indicate.



Source: LinkedIn, Job Openings and Labor Turnover Survey (JOLTS), Steven J. Davis, et al. HBR

Discussion Question 2

Analyzing Labor Market Indicators

According to the article, why is it important to look at multiple labor market indicators rather than relying solely on the unemployment rate?

Your Task

- 1 Identify at least two indicators from the article that tell different stories about the current labor market.
- 2 Explain what each indicator reveals about the state of employment in the U.S.

Think About It

"How might policymakers respond differently if they focused only on the unemployment rate versus looking at the full picture?"

How AI Could Reshape Jobs by 2030

World Economic Forum | January 7, 2026



Key Insight

The outcome depends on policy choices, corporate strategies, and how quickly workers can adapt to new technologies.

[Read the article here](#)

Scenario	AI Adoption	Labor Demand	Wages & Skills
Supercharged Progress	Rapid, transformative	New jobs emerge quickly	Wages rise with productivity. Requires AI fluency & creativity.
Age of Displacement	Fast, disruptive	Jobs lost faster than created	Wages decline for displaced. Urgent reskilling needed.
Co-Pilot Economy	Incremental, collaborative	Human-AI teams	Stable, gradual growth. Requires adaptability & AI tools.
Stalled Progress	Conservative, limited	Minimal change	Stagnant wages. Traditional skills remain dominant.

Discussion Question 3

AI and the Future of Work

Of the four scenarios presented in the article, which one do you think is most likely to occur, and why?

Your Task:

In your answer, consider how your chosen scenario would affect:

- Labor demand
- Wages
- The types of skills workers will need

Think About It

You may draw on examples from the article or your own observations about AI, in the way you use it, or you have seen other people use it.

SECTION 02

Measuring Unemployment and key labor market indicators

How Is the Labor Force Measured?

Bureau of Labor Statistics (US) | Planning and Statistics Authority (Qatar)

Every adult (16 and older) falls into one of three categories based on their employment status.

Category	Definition	Examples
Employed	Paid work (employees, self-employed) or unpaid family business; temporarily absent (vacation, illness)	Full-time worker, part-time employee, business owner
Unemployed	Not employed, available for work, actively sought employment in past 4 weeks	Job seeker, recently laid off worker
Not in Labor Force	Not employed and not seeking work	Full-time students, homemakers, retirees

Key Insight

The labor force includes only the employed and unemployed. Those not seeking work are excluded from unemployment calculations.

Three Essential Labor Market Measures

01 Labor Force (LF)

$$\text{LF} = \text{Employed} + \text{Unemployed}$$

02 Unemployment Rate (UR)

$$\text{UR} = (\text{Unemployed} / \text{Labor Force}) \times 100$$

03 Labor Force Participation Rate (LFPR)

$$\text{LFPR} = (\text{Labor Force} / \text{Adult Population}) \times 100$$

Key Insight

Among these three measures, which one is most telling of economic conditions? The unemployment rate is widely reported, but the employment-population ratio may provide a more complete picture.

ACTIVE LEARNING 01

Calculating Labor Market Indicators

Active Learning 1

Labor Force Calculations

Hypothetical Economy Data

Total Adult Population (16+)

260,000,000

Employed

150,000,000

Unemployed

10,000,000

Calculate the following:

- 1 Calculate the **Labor Force** .
- 2 Calculate the **Unemployment Rate** .
- 3 Calculate the **Labor Force Participation Rate** .
- 4 If 5 million unemployed workers become discouraged and stop looking for work, what happens to the unemployment rate?

Active Learning 1

Solution

Part 1: Initial Calculations

Measure	Formula	Calculation & Result
Labor Force	Employed + Unemployed	150M + 10M = 160,000,000
Unemployment Rate	(Unemployed / LF) × 100	(10M / 160M) × 100 = 6.25%
LFPR	(LF / Adult Pop) × 100	(160M / 260M) × 100 = 61.5%

Part 2: Discouraged Workers Scenario

New Unemployed	10M - 5M = 5M
New Labor Force	150M + 5M = 155M
New Unemployment Rate	(5M / 155M) × 100 = 3.2%
Key Takeaway The unemployment rate fell from 6.25% to 3.2%, but this does not reflect an improvement—5 million people simply stopped looking for work. This illustrates why we need multiple indicators.	

Challenges in Measuring Unemployment

Distinguishing between unemployed and out of the labor force is difficult. Several groups complicate measurement.

Group	Challenge	Impact on Statistics
Discouraged Workers	Stopped looking due to lack of opportunities	Unemployment rate understates joblessness
Marginally Attached	Want work but haven't searched recently	Not counted as unemployed
Part-Time for Economic Reasons	Want full-time but can only find part-time	Counted as employed despite underemployment
Underground Economy	Work "off the books" to avoid taxes	May report as unemployed

Key Insight

The official unemployment rate (U-3) may understate true labor market slack. The broader U-6 measure includes discouraged workers and involuntary part-time workers.

Duration of Unemployment Matters

Under 5 weeks	~33%
Under 14 weeks	~66%
Over 6 months	Only 21%

Important Paradox

"Most spells of unemployment are short, but most unemployment observed at any given time is long-term."

Key Insight

The small group of long-term unemployed has little turnover, so it accounts for most unemployment observed at any point in time.

Why This Matters

Long-term unemployment has severe economic and psychological consequences:

- **Skills Deterioration:**
Professional capabilities fade without use.
- **Reduced Future Earnings:**
"Scarring effect" on career trajectory.
- **Mental Health Impacts:**
Increased stress and loss of identity.
- **Social Stigma:**
Employers may view long gaps negatively.

SECTION 03

Types of Unemployment

Frictional, Structural, and Cyclical

Understanding the Natural Rate of Unemployment

Unemployment can be divided into categories based on its causes and duration.

Type	Duration	Cause	Example
Frictional	Short-term, voluntary	Time to match workers with jobs	Recent graduate searching for first job
Structural	Long-term, involuntary	Mismatch between skills and job requirements	Coal miner in a region shifting to renewable energy
Cyclical	Varies with business cycle	Economic downturns reduce demand	Workers laid off during recession

Natural Rate Formula

Natural Rate = Frictional + Structural Unemployment

Key Insight

"Natural" does not mean desirable. It represents the unemployment rate when the economy is at full employment—cyclical unemployment is zero.

Frictional Unemployment: The Search Process

Unemployment that results from the time it takes to match workers with jobs. **It is often** short-term and voluntary.

Why It Exists

- **Imperfect Information:**
Workers don't know about every opening; firms don't know every candidate.
- **Sectoral Shifts:**
Changes in the composition of demand among industries or regions.
- **Career Transitions:**
Workers leaving jobs to find better matches.



Frictional unemployment ends when a successful match is made.

Is It Bad?

Not necessarily. Some frictional unemployment is inevitable and even beneficial. It indicates workers are taking time to find jobs that best suit their skills, leading to higher productivity and wages.

Structural Unemployment: Skills Mismatch

Occurs when there is a fundamental mismatch between the skills workers possess and the requirements of available jobs.

Primary Causes

Technological Change

Automation and AI replace routine cognitive and manual tasks, making certain skills obsolete.

Globalization

Production of goods moves to countries with lower labor costs, displacing domestic workers.

Sectoral Shifts

Permanent changes in consumer demand (e.g., the shift from coal to renewable energy).



Automation increases productivity but can displace workers with specific skill sets.

Key Difference: Unlike frictional unemployment, structural unemployment often lasts longer because workers need time to retrain or relocate.